

Research Statement

Adaptive and Reliable Learning in AI Agents

I am interested in how AI agents learn from accumulated experience and remain reliable as tasks, environments, and user needs change. My current work approaches this problem from two complementary perspectives: how agents represent and reuse experience across tasks, and how state representations and evaluative feedback guide decisions within long-horizon interactions. More broadly, I am interested in how memories, representations, and evaluators should be revised as the agent, its environment, or the objectives used to assess it evolve. My work in continual learning and an internship on automated research workflows provide the starting points for this agenda.

My previous research, *Stability-Plasticity Theory for Continual Learning under Drifting Task Optima*, develops a simplified theoretical model in which past prediction errors shape a history-dependent update metric. The analysis provides a tracking guarantee for drifting task optima and relates retention to current-task error and movement between optima. A controlled three-task construction fixes the pairwise distances between task optima and isolates an order-dependent retention effect that disappears when metric learning is disabled. Although this model does not establish the same mechanism in deep networks, it shaped my broader interest in how learning history affects subsequent adaptation. I now want to study this question in AI agents whose behavior is influenced by external memories, learned representations, and feedback accumulated over extended interactions.

Learning reusable memories from experience

I want to understand how agents should extract, organize, revise, and learn from interaction histories. I am particularly interested in the relationship between episodic memory, which retains specific trajectories and their context, and procedural memory, which abstracts reusable strategies. The challenge is to preserve information that supports future adaptation under limited storage and computation as tasks, tools, environments, and the agent’s own capabilities change.

Recent work learns memory-management policies, estimates memory utility from feedback, and couples experience extraction with actor updates [1, 2, 3]. Building on these approaches, I want to study memory utility as a conditional quantity: a memory’s marginal contribution depends on the current policy, other stored memories, and the distribution of future tasks. This raises a temporal credit-assignment question: how should an agent evaluate a memory-writing or consolidation decision whose benefit emerges only during later tasks or policy updates?

One starting point is how abstraction changes a memory’s applicability. A successful workaround may depend on a particular tool or assumption; distilling it into a general instruction can remove those conditions. I would compare raw episodes, distilled strategies, and memories that retain supporting evidence and applicability conditions. Controlled removal or replacement of memories could help measure their marginal effects and interactions. Under matched storage, token, and tool-use budgets, I would evaluate transfer to held-out task families, adaptation efficiency, and negative transfer after environmental changes. Comparing a frozen actor with a continually updated actor would also help distinguish adaptation through external memory from the contribution of stored experience to subsequent parameter learning.

State representation and reliable feedback

During my internship on automated research workflows, I encountered cases where a failed replication or diagnostic check could improve the next decision, while a sequence that appeared productive left the final evidence no stronger. This led to an emerging interest in feedback for agents that must decide whether to explore, verify, repeat, switch direction, ask for assistance, or stop.

Existing approaches include process advantages based on future answer success and generative step verification [4, 5]. Learned internal states support memory consolidation [6], while environment-aware verifiers detect hidden errors and distinguish recoverable exploration from other failures [7]. These developments motivate me to examine how state representation, evaluation targets, and supervision jointly affect the reliability of feedback.

More generally, the objective used to evaluate an agent may itself be only partially observed or change over time. In personalized or human-facing systems, for example, the relevant utility can depend on user goals that vary across tasks and gradually evolve through interaction. Recent work on personalized deep research shows that synthetic-user evaluation can miss failures revealed through direct interaction with real users [8]. An evaluator must therefore distinguish genuine changes in objectives from temporary context or noisy feedback, adapting when necessary without overwriting stable preferences. I view this as another form of the stability-plasticity problem: not only the agent, but also the mechanism evaluating it must remain both stable and responsive.

I view a long research workflow as a partially observed decision process. An agent compresses its interaction history h_t and relevant memory M into a state $z_t = f_\phi(h_t, M)$ for action selection and evaluation. I want to investigate whether this representation preserves the distinctions needed to rank subsequent actions. For example, two failed experiments may look similar, yet one rules out a relevant explanation while the other repeats an uninformative test. Merging these histories can cause state aliasing: the same representation conceals different evidence and decision values. Candidate representations would track verified claims, unresolved dependencies, uncertainty, and remaining resources.

I am also interested in distinguishing local validity from an action’s expected contribution to final utility. A critic $Q^\pi(z_t, a_t)$ can estimate expected utility under a continuation policy, incorporating evidence validity, interaction cost, and the option value of gathering more information; an advantage target then compares that value with the policy’s baseline at the same state. Even binary terminal labels can support continuous value estimates, but their meaning depends on which utility is being estimated, what information the critic observes, and whether the target remains appropriate as the policy or user objective changes.

Controlled experiment-selection environments would let me vary state compression separately from the feedback target and its supervision. I would compare outcome-derived value estimates with signals augmented by tool-based or human verification, using reference action values where tractable. Evaluation would test action rankings, stopping and escalation decisions, and independently verified outcomes under matched total budgets. In longitudinal settings, I would additionally measure adaptation delay after a genuine change in objectives, false updates caused by temporary context, retention of stable preferences, and the ability to recognize when direct feedback is needed. These experiments could distinguish failures caused by information loss, target misspecification, preference-model error, or value-estimation error before using the feedback for search or reinforcement learning.

These directions concern how experience is represented, evaluated, and reused at multiple timescales: the state that supports decisions within an interaction, the memories that support learning across

tasks, and the evaluative objectives that may evolve with the environment or user. Continual learning provides my established research foundation, while agent memory, reliable feedback, and adaptive evaluation are directions I hope to develop during a PhD. I aim to combine theoretical analysis, controlled experiments, and evaluations of working agents to understand when accumulated experience leads to reliable and transferable improvement.

References

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